IRRIGATION WATER MANAGEMENT · NOTE WM06

# Underground Pipeline Distribution — Capacity, Structures and PVC Loop Systems

*Course topics: IDE5.5 L8, CAE1.3 L13, IDE-503 T9*

## Why underground pipelines

- They carry and distribute farm water with practically no conveyance loss; the land above stays under crops and fields are not cut into short runs (open channels may take 2–4% of the land).
- They are essentially leak-proof, protected below ground, work under pressure so can run uphill or downhill, reach areas open channels cannot, and are not clogged by weeds or wind-blown material.
- Pumped well water can go straight into the pipeline, so the well need not be at the highest point; place outlets for the best supply.
- They cost more than lined channels but pay where fields lie at different levels and the land saved is counted. They generally do not suit canal irrigation (too little head, silty water) but suit pumping schemes, needing very little extra power.
- **Pipes:** factory-made reinforced concrete or PVC; non-reinforced concrete for small low-pressure farms.

## Discharge capacity (Darcy)

$$v = \sqrt{[\, H\, d\, g\, /\, (2\, f\, l)\, ]}$$

v = velocity (cm/s); H = available head — the difference between the water level in the pump stand and the outlet (cm); d = pipe diameter (cm); g = 981 cm/s²; l = pipe length (cm); f = Darcy roughness coefficient (tabulated for different pipes and velocities).

> Example 7.7: d = 15 cm, length 200 m, elevation difference 2.5 m, f = 0.009. v = √(250 × 15 × 981 / (2 × 0.009 × 20,000)) ≈ 102 cm/s = 1.02 m/s. Area = 0.0176 sq m, so Q = a v = 0.0179 cu m/sec = 17.9 litres/sec.

## Pipeline structures

| Structure | Purpose and details |
|---|---|
| Inlet — pump stand | The commonest inlet: a vertical pipe above ground, larger than the pipeline (desirably 75 cm diameter), tall enough not to overflow except under unusual pressure. Its water level must overcome friction and valve-pressure losses. Every inlet must develop full capacity, hold enough head, prevent excess pressure and keep out trash |
| Air vents | Vertical pipes that release trapped air and prevent vacuum: 80–100 cm downstream of the pump stand, at all high points and sharp turns, at abrupt grade changes below air-entraining structures and at the end of the line; about every 150 m on straight uniform slopes. A 5 cm steel pipe on a 60 cm length of concrete pipe at least ½ the main diameter |
| Gate stands | Like pump stands, where the line branches; slide gates or valves control flow into the laterals |
| End plug | Closes the line where it ends and absorbs water-hammer pressure; a concrete cap backed by a masonry wall |
| Riser valves | Outlets about every 60 m: a riser pipe saddled over a hole cut in the top of the line with a permanent cement-mortar joint, topped by a screw valve with a rubber-sealed cap plate; raising or lowering the plate controls the flow |
| Hydrants | Portable fittings over riser valves to connect gated pipes; moved as irrigation proceeds; the riser valve stays adjustable |
| Gated pipes | Light portable aluminium or steel pipes with evenly spaced finger-operated rubber or metal slide gates, feeding furrows, borders and check basins with minimal seepage loss |

## PVC pipeline systems (deep tube-well commands)

- PVC pipes are increasingly used for on-farm distribution and deep tube-well irrigation: easy to install, leak-proof joints, fast laying and high resistance to internal pressure and surges — but dearer than concrete in large diameters carrying large flows. Survey, layout and diameter design are the same as for any pipe; the structures are nearly identical except the intake.
- **Pipe standard:** IS: 4985-1981 (as amended) or other national standards; plain-end pipes welded (solvent-cemented) into leak-proof joints; standard fittings (Fig. 7.42): coupling, cap, plug, adaptor, tee, 45° elbow and 90° elbow.

> **Terminology fix.** The notes call these "plasticised" PVC pipes. IS 4985 covers **unplasticised** (rigid) PVC — uPVC — pipes for water supply; the standard has since been revised, so check the current edition.

| Layout | Where | Feature |
|---|---|---|
| Branching network | Undulating terrain | Needs a greater length of pipe |
| Looped network | Plain areas — suitable and economical | Pipe diameter about 30% lower than a branching system for the same capacity |
| Closed-loop system | Usually two loops, one on either side of the well | Only one valve open per loop at a time; about 40 m of pipe per hectare served; each loop commands up to 50 ha |

### Key components

- **Elevated tank** near the tube well acts as the pump stand; its water level sets the line pressure. Two compartments (one per loop) with a 30 cm stand pipe that releases pressure and prevents water

hammer. No separate air vents or vacuum valves are needed: delivery valves release air during first filling, the line stays full when the pump stops, outlet compartments act as vents and air bubbles are seldom carried down.

- **Surge riser** (one per loop, at the highest point near the middle of the loop) protects against spill-over surges and water hammer on valve closure: a 20 cm RCC pipe with a perforated PVC cap, its base in masonry, its top about 50 cm above the hydraulic grade line so water does not spill.
- **Laying:** about 1 m deep; the line is designed to stay full (no gravity drainage), so water comes at once when the pump restarts. To empty it, remove the head cover of the lowest outlet valve and pump out with a portable centrifugal or positive-displacement pump.
- **Outlet valves:** each commands up to 5 ha on plain land; place them on the boundary between two or more holdings, near the centre of the command. Use conventional alfalfa-type riser valves of the riser-pipe size (usually 15–20 cm) fixed on T-joints, embedded with part of the riser in masonry or concrete, with gated distribution boxes.
- **Testing:** backfill the trenches and build the outlets; run the tube well to fill the line (air escapes through the pressure-release pipe); for 3–4 days keep all outlets open 3–4 hours daily to check for leaks; drain the whole system before any repair.

**Sources:** ICAR eCourse 'Irrigation Engineering' — Dr K. N. Tiwari and Dr N. S. Raghuwanshi (IIT Kharagpur), reviewed by Dr S. D. Gorantiwar (MPKV Rahuri). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

## Practice MCQs

**Q1. Open channels may occupy about ___ of farm land, saved by pipelines**

(a) 0.2–0.4% (b) 2–4%
(c) 20–40% (d) 50%

**Q2. The most common inlet structure of an underground pipeline is the**

(a) gate stand (b) pump stand
(c) end plug (d) hydrant

**Q3. Air vents on straight uniform slopes are spaced about**

(a) 15 m (b) 60 m
(c) 150 m (d) 1 km

**Q4. Riser valves are usually spaced about**

(a) 6 m (b) 60 m
(c) 600 m (d) 1.5 km

**Q5. In a closed-loop PVC system, each loop may command up to**

(a) 5 ha (b) 20 ha
(c) 50 ha (d) 500 ha

**Q6. IS 4985 covers**

(a) plasticised PVC (b) unplasticised (rigid) PVC pipes
(c) HDPE drip laterals (d) RCC pipes

**Answers**

Q1 — (b) 2–4%
Q2 — (b) pump stand
Q3 — (c) 150 m
Q4 — (b) 60 m

Q5 — (c) 50 ha

Q6 — (b) unplasticised (rigid) PVC pipes